

Mutual truthfulness and known mechanicalness: a proof of Alexander’s Question 3-1-3

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Abstract

Alexander’s Question 3-1-3 asks whether a family of knowing machines can know everyone’s truthfulness, know particular codes for its subordinates, and know everyone’s mechanicalness. We give a proposed affirmative proof for every recursively enumerable well-founded relation of subordination, with no restriction on the formulas. A fixed common theory supplies mechanicalness; the original machines have particular codes for this theory and their direct subordinates. To prove the resulting family sound, we use ordinal labels, auxiliary modal symbols that remember their original owners, and finite hereditary restrictions of external knowledge. Guarded transport recovers full source proof predicates at prescribed labels. A separate core argument and an induction over whole downward ideals then establish soundness for arbitrary hierarchies. We extract the required effective ordinal structure from Wilken’s pure order-two resemblance analysis, including the domains of its partial base conversions. The final reduct has precisely Alexander’s original operators. This paper presents the complete proposed hand argument; its separate Lean formalization is unfinished. No literature priority is claimed.

1 The question and the affirmative statement

A machine may know that another machine has some program without knowing which program it has. The distinction is the position of the existential quantifier. For a formula ϕ with lone free variable x , the two assertions are

$$\exists e K_i \forall x (K_j \phi \leftrightarrow x \in W_e), \quad K_i \exists e \forall x (K_j \phi \leftrightarrow x \in W_e).$$

The first supplies a particular index that i knows to describe j ’s column. The second says that i knows such an index exists. Here W_e is the e th recursively enumerable subset of $\mathbb{N}$, and $x \in W_e$ abbreviates its standard arithmetical enumeration formula.

Let $\mathcal{L}$ be the language of Peano arithmetic with operators $K_0, K_1, \dots$. We use Alexander’s weak modal semantics [1, §§1.2–1.3]. Knowledge is a formula-forming operator; it is not given Kripke accessibility semantics. The arithmetic part of our final structure is the standard natural numbers.

A family of theories $i \mapsto T_i$ determines knowledge by

$$M \models K_i \theta[s] \iff T_i \vdash \theta^s. \tag{1}$$

The sentence θ^s replaces each free variable by the numeral for its value under s . Throughout, $\vdash$ means consequence in Alexander’s base logic. Completeness makes it equivalent to an effective

finite-certificate proof relation. A family of knowing machines has recursively enumerable theories and satisfies the self-doubting knowledge axioms: validity, deduction, introspection and truthfulness, together with knowledge of the first three. Self-doubting Epistemic Arithmetic additionally includes modal PA and knowledge of its axioms. The induction schema ranges over the whole modal language.

Write jRi when j is subordinate to i . We assume R is recursively enumerable and well-founded. It need not be transitive. Let R^+ be its positive transitive closure; a path of length zero is allowed when we speak of a downward ideal.

Theorem 1 (full 3-1-3). For every recursively enumerable well-founded relation R on $\mathbb{N}$, there is a uniformly recursively enumerable family of knowing machines satisfying self-doubting Epistemic Arithmetic and the following schemas in $\mathcal{L}$:

1. For all i, j and every formula θ , $K_i \text{ ucl}(K_j \theta \rightarrow \theta)$.
2. For jRi and every ϕ with $\text{FV}(\phi) = \{x\}$, $\exists e K_i \forall x (K_j \phi \leftrightarrow x \in W_e)$.
3. For all i, j and every ϕ with $\text{FV}(\phi) = \{x\}$, $K_i \exists e \forall x (K_j \phi \leftrightarrow x \in W_e)$.

These are the three entries defining Question 3-1-3 in [1, Definition 1.1.1]. All formulas, including formulas about superior and incomparable agents, are included in one family. There is no bound on modal depth, arithmetical complexity or hierarchy height. In particular, no recursive rank function for R is assumed.

Corollary 2. The same family gives affirmative answers to Questions 3-1-1 and 3-1-2.

Indeed, their mechanicalness requirements are subcollections of the third schema. We therefore prove the stronger statement directly.

How the proof is organized

The main construction is short. We add a temporary common knower $*$, put all existence-of-code axioms in its theory U , and give each original theory T_i particular codes for U and its subordinates. Every T_i can then inherit the closed theorems of U . What requires work is proving that U remains sound when all these code-holding machines are present.

We first separate proof membership from truth by filtering each raw proof predicate through the truth of its argument. This makes a structure before soundness is known. Ordinal stages let us verify its generators using smaller cuts. External modal occurrences that would otherwise force a circular soundness assumption are restricted to finitely many hereditary argument shapes. Auxiliary symbols record the owners of discarded occurrences, so that arithmetic program indices never have to change. The resulting local reconstruction permits an induction on the original hierarchy. The two measures serve different purposes: ordinal labels control active proof cuts; original hierarchy ranks control the lower theories retained in a recursive call.

Sections 2–6 give the construction and its local lemmas. Sections 7–9 establish the global result. Section 10 proves the precise ordinal input from published results and effective procedures, so the main argument has no additional ordinal hypothesis.

2 Fixed theories, fixed programs, and the truth filter

Work temporarily in $\mathcal{L}^* = \mathcal{L} \cup \{K_*\}$. For a holder $a \in \{*\} \cup \mathbb{N}$, let B_a contain the following closed generators: modal PA; Assigned Validity; a -Validity, a -Deduction and a -Introspection; and $\text{ucl}(K_b\theta \rightarrow \theta)$ for every b, θ . Assigned Validity means η^s for every base-logically valid η and numerical assignment s . All schematic formulas are universally closed where appropriate.

If Γ is a set of generating sentences, $\text{cl}_a(\Gamma)$ contains $K_a^n\sigma$ for $\sigma \in \Gamma$ and $n \geq 0$, where the superscript here denotes repeated application. We distinguish this ordinary finite prefix from the ordinal-labelled heads introduced below. Write $\text{Th}(S) = \{\sigma : S \vdash \sigma\}$.

For a fresh index variable e , put

$$\text{MC}_b(\phi) = \text{ucl} \exists e \forall x (K_b\phi \leftrightarrow x \in W_e).$$

Other free parameters of ϕ are closed *outside* the existential: an index may depend on their values. Define

$$U = \text{cl}_*(B_* \cup \{\text{MC}_b(\phi) : b, \phi\}). \quad (2)$$

This recursively enumerable theory is fixed independently of R . It has no particular-code generators. Choose an effective compiler d for all its unary consequence columns:

$$n \in W_{d(\phi)} \iff U \vdash \phi(\bar{n}), \quad \text{FV}(\phi) = \{x\}. \quad (3)$$

Given a provisional family index q , let $c(q, j, \phi)$ be the program enumerating the ϕ -column of the theory with owner j enumerated by q . The parameter theorem makes c effective. Define

$$\begin{aligned} T_i(q) = \text{cl}_i(B_i \cup \{ \forall x (K_*\phi \leftrightarrow x \in W_{d(\phi)}) : \text{FV}(\phi) = \{x\} \} \\ \cup \{ \forall x (K_j\phi \leftrightarrow x \in W_{c(q,j,\phi)}) : jRi, \text{FV}(\phi) = \{x\} \}). \end{aligned} \quad (4)$$

The family is uniformly recursively enumerable in q : every edge is admitted on a positive enumeration witness. The recursion theorem chooses q for this family. Set $T_i = T_i(q)$. Then

$$n \in W_{c(q,j,\phi)} \iff T_j \vdash \phi(\bar{n}). \quad (5)$$

Equations (3) and (5) are external identities of enumerations. We never add an axiom saying that a theory knows its own enumerator identity. The theories, q and all displayed numerals are fixed from now on.

Lemma 3 (own-known finite consequences). If $S = \text{cl}_a(\Gamma)$ contains own Validity and Deduction, then $S \vdash \sigma$ for a sentence σ implies $S \vdash K_a\sigma$.

Proof. A consequence certificate uses finitely many generating premises $\gamma_1, \dots, \gamma_m$ and a valid implication from them to σ . Their own prefixes belong to S . Own Validity gives knowledge of the implication, and repeated own Deduction gives $K_a\sigma$. The same argument iterates. It is a derived rule for sentences from finite certificates, rather than an added global necessitation rule. $\square$

For raw predicates P_a on closed formulas, define F_P recursively by standard arithmetic and

$$F_P \models K_a\theta[s] \iff P_a(\theta^s) \text{ and } F_P \models \theta[s]. \quad (6)$$

We call a raw family admissible when it is uniformly recursively enumerable and invariant under alphabetic variation. We call P_a *fully sound in F_P* when $P_a(\sigma)$ implies $F_P \models \sigma$ for every sentence σ . This refers to every accepted sentence, not merely to assertions currently being inspected.

Lemma 4 (filtered structures). An admissible raw family defines a legitimate Alexander structure F_P . Numerical assignment commutes with truth: $F_P \models \theta[s]$ iff $F_P \models \theta^s$. The structure satisfies modal PA and every reflection instance $K_a \theta \rightarrow \theta$.

Proof. The truth recursion in (6) calls only a proper subformula. Alphabetic invariance gives alphabetic invariance of truth. Dependence only on free variables and the permitted variable-for-variable substitution condition follow because numerical closure supplies the same sentence on both sides, up to alphabetic variation. Formula induction gives the assignment identity. Reflection is immediate from the truth conjunct. For modal induction, the sentences at $x = \bar{n}$ and $x = S(\bar{n})$ give the ordinary induction on n in the standard natural numbers. Here $S(\bar{n})$ is literally $\bar{n} + 1$. This is Alexander's argument in Lemma 1.2.22 and Proposition 1.2.23, and uses no arbitrary-term substitution law inside knowledge. $\square$

The actual raw family will be $P_* = \text{Th}(U)$ and $P_i = \text{Th}(T_i)$. Proving its full soundness removes every truth conjunct in (6) and yields (1). Until then, an enumerated proof alone does not verify a particular-code axiom.

3 The ordinal room used by finite proofs

We state the exact label facts now and derive them in Section 10. Let L be a recursive notation domain for the well-order $[0, v_\omega)$, carrying recursive relations $<, \leq_1, \leq_2$ from Wilken's *pure* order-two structure. Write $<_i$ for a strict comparison. The required properties are:

1. $\leq_2 \subseteq \leq_1 \subseteq \leq$; the resemblance orders are transitive. If $p \leq_1 r$ and $p < q \leq r$, then $p \leq_1 q$.
2. If $\alpha \leq_1 \beta$, a finite expanded diagram below β can be reflected below α , fixing its already prescribed finite part below α . The language includes both resemblance relations, and the diagram includes their negative facts.
3. A guard $G_k(\lambda)$ is witnessed by $\lambda = g_0 <_2 g_1 <_2 \dots <_2 g_k$. If $k \geq 1$, then above any finite support $X < \lambda$ and any bound $u < \lambda$, there are arbitrarily many finite copies $Y_0 < \dots < Y_{m-1} < \lambda$ of the first k guard points. Every copied point is $\leq_1$ -related to λ . Thus $\max Y_t < \min Y_{t+1}$: separation concerns whole copies.
4. There is a cofinal strict chain $C = \{v_n : n \geq 2\}$ with $v_2 <_2 v_3 <_2 \dots$.

The enumerators use finite relation checks and finite witnessed guards. They need no decision procedure for membership in C . A G_0 guard is one point. The room statement is only used with $k \geq 1$.

There are two kinds of induction in this paper. The well-order L supports effective labels for finite auxiliary proofs. The rank of R will be used only as a metatheoretic measure; it never selects axioms.

4 Recorded formulas and auxiliary presentations

Fix a finite active set A of holders and a finite set E of old external original names. There are two applications:

- In a *core presentation*, $A = \{*\}$, the source holder is U , and $E \subseteq \mathbb{N}$ is arbitrary. Main generators are the mechanicalness generators of U .

- In an *extension presentation*, $A = J$ is a finite R^+ -antichain, the source holder at $a \in J$ is T_a , and E is disjoint from the whole original downward ideal $D_R(J)$. Main generators are the particular core and subordinate codes of T_a . There are no mechanicalness generators in this presentation.

For $e \in E$ introduce a fresh free modal symbol Z_e . An active head may be labelled K_a^ν , for $\nu \in L$. The *source erasure* ε removes ordinal labels, sends Z_e to K_e , fixes every other head and fixes arithmetic. Thus a ghost occurrence still remembers its original owner. The adjective “ghost” describes the auxiliary model; Z_e is an ordinary unconstrained modal symbol in logical certificates.

For example, an auxiliary version of a source code on $K_e\tau$ may display $Z_e\tau$, but its program numeral is still the numeral for the source argument $K_e\tau$. It is never the program for a different original operator K_s . Erasure keeps the code identity while allowing the auxiliary model to make the unwanted occurrence false.

Define the skeleton $\text{sk}(\theta)$ by preserving formula constructors, arithmetic predicate symbols and modal owners, while replacing terms and binder variable identities by wildcards. Let $r(\theta)$ be its constructor count. Proper formula subexpressions have smaller rank. Alphabetic variation, numerical assignment and permitted variable substitution preserve skeletons, including when two variables merge.

A finite hereditary protection table H consists of pairs $(e, \text{sk}(\chi))$. The map π_H keeps $K_e\chi$ live if its pair is in H , and otherwise changes its head to Z_e ; it applies recursively to arguments. Heredity requires that a retained external argument retain all its nested external arguments. Consequently

$$\varepsilon\pi_H\theta = \theta, \quad \text{every retained external argument has a } Z\text{-free subtree.} \quad (7)$$

A source-shared labelling assigns the same label to equal pairs $(a, \text{sk}(\varepsilon\chi))$. Every active label lies strictly above *all* active labels anywhere in its argument, also through live unindexed or ghost contexts. A formula satisfying this nesting is globally admissible. Main mechanicalness and code images use one protection table and one source-shared label table. Structural and validity premises require global admissibility, without a requirement that unrelated premises share a label table.

Lemma 5 (logical maps). Erasure, hereditary protection and a global active stratifier preserve base-logical validity. They preserve free variables, alphabetic variation and permitted substitutions, and commute with numerical closure. Their finite certificate translations are effective.

Proof. In Alexander’s first-order encoding Δ , a modal atom $O\chi$ is a separate predicate applied to the ordered tuple of its free variables. The axioms BL1 and BL2 enforce alphabetic variation and variable substitution. Each map sends this predicate to that of its mapped modal atom, with the same free variables. Its images of BL1 and BL2 are again instances of those schemes. Renaming or identifying predicates preserves a first-order validity certificate. Alternatively, pull back an arbitrary target structure along the map and apply formula induction; the substitution identities make the pullback a legitimate weak-modal structure.

For a global stratifier, prescribe a consistent finite table first. Process argument shapes in increasing rank, choosing each new label above the finitely many labels of its descendants. Search in the recursive unbounded order finds such a label. Repeated source shapes receive the same entry. Thus the stratifier commutes with the required substitutions. Numerical closure identities follow by recursion.

Only the stated direction of erasure is used. The converse would be false: $K_e\tau \leftrightarrow Z_e\tau$ erases to a tautology but is not a base-logical validity. $\square$

The generator rules

For each active holder a enumerate a theory Λ_a . All generators are closed sentences, with numerical instances admitted as in the source. The following table specifies the structural part.

Class	Admitted auxiliary generator
Arithmetic	Globally admissible modal PA, including full modal induction.
Assigned validity	η^s for valid globally admissible η .
Own validity	$K_a^\nu\eta$ for valid η with argument support below ν .
Own deduction	Same-label deduction with every argument support below the label.
Own introspection	$K_a^\mu\eta \rightarrow K_a^\nu K_a^\mu\eta$, where $\mu < \nu$ and the whole formula is globally admissible.
Reflection	Truthfulness of indexed, unindexed and ghost heads.
Contextual collapse	Equivalence when one active label is changed along $\leq_1$ inside a globally admissible context.
Own prefixes	Prefix any generating sentence at a holder label above its entire support, and iterate.

For contextual collapse, the changed argument lies below both labels; all intermediate contexts used are globally admissible. It is available to each holder. It is not an import of another holder's structural axioms.

The main generators are labelled π_H -images of the source mechanicalness sentences (core type) or particular codes (extension type). In an extension image the displayed code is

$$\forall x(K_w l\pi_H\phi \leftrightarrow x \in W_{p(w,\phi)}), \quad p(w,\phi) = \begin{cases} d(\phi), & w = *, \\ c(q, w, \phi), & w \in \mathbb{N}. \end{cases} \quad (8)$$

Its owner w is neither active nor old external. Its numerator uses the *source* ϕ , not the visible formula $l\pi_H\phi$.

Inspect the whole distinguished modal atom of a main image, including its outer head. Whenever $K_e\chi$ is live with $e \in E$, every active atom $K_b^\nu\beta$ inside χ must have a finite witness for $G_{r(\varepsilon\chi)}(\nu)$. Its argument support must be below ν and $\leq_1$ -related to ν . Inspect nested live heads even below ghosts. In particular, a core mechanicalness target $K_e\chi$ imposes guards on the active atoms of its argument even when there is no other external head in χ .

Every finite guard point h is made actual formula support by padding the main sentence with an implication from holder-local tautology markers $K_a^h\tau_h$. Choose distinct arithmetic tautology shapes absent from the main formula. The tags are freely chosen formulas; they do not encode the notation number of h . A reflected label keeps the same tag. Each marker is provable by the holder's own validity. The padded image therefore yields the unpadded main sentence by discharging its markers.

All admission checks are finite and recursive on label notations. Source validity and generator certificates are recursively enumerable. For extensions, positive R -edge witnesses supply the sub-

ordinate codes. The antichain and ideal-separation hypotheses apply to the semantic uses of these theories; the enumeration makes no negative R -test and consults no hierarchy rank.

5 Erasure, collapse, and transport at a fixed label

Let S_a be the source theory, either U or T_a . The support of a recorded formula is the finite set of its active labels, including the marker labels. Let $\Lambda_a^{<\nu}$ contain exactly the generated sentences whose *whole* support is below ν , including those with empty support. A finite consequence certificate contains its premises, generator witnesses and a valid implication to its conclusion. This follows from Alexander's compactness and completeness theorem [1, Theorem 1.2.6 and §1.3].

Lemma 6 (source erasure). If $\Lambda_a \vdash \eta$, then $S_a \vdash \varepsilon\eta$. The same holds for every cut.

Proof. Each structural generator erases to its source counterpart; contextual collapse erases to the equivalence of a formula with itself. A padded main image erases to an implication from own-known arithmetic tautologies to the identical source mechanicalness or code generator. It is therefore a source consequence. A generated prefix erases to a source consequence by Lemma 3, including prefixes of discharged code images. Translate the final validity certificate by Lemma 5. The source argument and its literal program numeral are unchanged. $\square$

Lemma 7 (finite-support collapse). If $\alpha \leq_1 \beta$ and $\text{supp}(\eta) < \alpha$, then

$$\Lambda_a^{<\alpha} \vdash \eta \iff \Lambda_a^{<\beta} \vdash \eta. \quad (9)$$

Proof. The forward implication is inclusion. For the reverse, take the support of a finite proof below β . Reflect its expanded finite diagram below α , fixing every support point already below α , in particular the target's labels. The reflection preserves strict nesting, all guard witnesses, and each relation comparison in a contextual-collapse certificate. Owners, source formulas, protection tables, variable data and program numerals remain fixed. Logical certificates are renamed along the same finite label map. Every reflected premise is a generator in $\Lambda_a^{<\alpha}$, with conclusion η . $\square$

Lemma 8 (prefixing a derived sentence). If $\Lambda_a^{<\nu} \vdash \eta$ and $K_a^\nu \eta$ is globally admissible, then $\Lambda_a \vdash K_a^\nu \eta$ by a proof whose support is at most ν . It consequently belongs to every later cut.

Proof. Take a finite certificate for η . Prefix its generated premises at ν and use own validity on the certificate's valid implication. Repeated same-label deduction gives $K_a^\nu \eta$. All body support was strictly below ν . This also prefixes derived main images after their markers have been discharged. $\square$

Two coherent lifts of one source formula on a common $\leq_1$ -chain, with the same live/ghost modes, are equivalent in every cut containing their support. To see this, compare each label with the pointwise minimum of the two assignments. These minima preserve strict nesting: if each parent's label exceeds its child's in both tables, the same is true of their minima. Change children before parents using contextual collapse, keeping every context admissible. This observation lets a proof choose convenient canonical depth labels while preserving the prescribed target.

Lemma 9 (guarded fixed-label transport). Let β' be a Z -free source-shared lift of β with support below λ . Assume a witnessed $G_k(\lambda)$, $r(\beta) < k$, and that all target support points are $\leq_1$ -related to λ . Then, for every numerical assignment s ,

$$\Lambda_a^{<\lambda} \vdash (\beta')^s \iff S_a \vdash \beta^s. \quad (10)$$

The same fixed label, mode table and target table work for every term variant of the prescribed source skeleton and every assignment.

Proof. Forward transport is Lemma 6. For reverse transport choose a finite source certificate for β^s . Protect exactly the external argument shapes of β , including their hereditary closure, and apply π_H to the certificate. Its target stays Z -free and unchanged. Every live external argument in a transformed generating premise has one of these protected shapes. It is a proper subexpression shape of β , hence has rank less than $r(\beta) < k$. External arguments of other shapes become ghosts, regardless of how deeply they occur.

Count active nesting through all heads, including unindexed and ghost heads. The finite transformed certificate has some finite maximum depth. Use the room property to choose enough whole-copy-separated G_{k-1} copies $Y_0, \dots, Y_N$ below λ , above the fixed target support. Put $c_t = \min Y_t$. Every copied and target point is $\leq_1$ -related to λ . Connectedness therefore makes their ordered union a single $\leq_1$ -chain. Label an active occurrence of depth t by c_t . A guard of rank $r \leq k-1$ uses the first $r+1$ points of Y_t .

We verify each premise type. PA and Assigned Validity commute with the logical maps. Own validity prefixes the lifted valid argument. Same-label deduction can be used at canonical conclusion labels, and its result can be adjusted along the common chain by contextual collapse. Introspection uses successive depth labels. Reflection becomes live, ghost or indexed reflection as appropriate. A source main generator has exactly its admitted protected image, with the original source numeral. All its live external arguments have the rank bounds just proved, so its guards fit; discharge its holder-local markers. For an own prefix, the proof of its body, including every point of every earlier guard copy, precedes the next depth base. Lemma 8 supplies the required prefix. Whole-copy separation is what justifies this last step; separation of the bases alone would not do so.

Translate the finite valid implication. Its canonical target is equivalent to the prescribed β' on the common chain, so we obtain the prescribed target below λ . The target and cutoff have never moved. Different s can require source proofs of different depths and hence different numbers of room copies. The room property allows every finite number; no uniform depth bound is required.

Finally, a term variant has the same protection classes, ranks and target labels. Apply the transformation to its actual finite source certificate, preserving its own source program numerals. This gives the statement for all variants, not only for the initially written formula. $\square$

Lemma 10 (cofinal recovery). For every globally admissible, source-shared, Z -free lift η whose labels are on C ,

$$\Lambda_a \vdash \eta \iff S_a \vdash \varepsilon\eta. \quad (11)$$

At a head labelled $\nu \in C$, its argument's cut test already equals the full source test below that same ν , for all assignments and term variants.

Proof. Place the preceding finite construction on a sufficiently long part of C . All positive nesting and guard requirements fit, and the canonical target is equivalent to the given one. To identify a test below a prescribed ν , first obtain the argument proof below a later C -point κ . Since $\nu \leq_1 \kappa$ and the argument support is below ν , Lemma 7 collapses it to $\Lambda_a^{<\nu}$. Only finite positive patterns are placed on C ; arbitrary negative diagrams are reflected in the ambient label structure instead. $\square$

6 The staged model and fully sound reconstruction

The staged model M has standard arithmetic. Its raw tests are:

- at a live unindexed head K_d , the original predicate $P_d((\varepsilon\eta)^s)$ on its argument η ;
- at an active head K_a^ν , the cut predicate $\Lambda_a^{<\nu} \vdash \eta^s$;
- at Z_e , the empty predicate.

Every test is filtered by the staged truth of its argument as in (6). Unmentioned original heads will have empty tests except for those explicitly retained in full. The unindexed test uses *source erasure*; it does not compile the formula visibly containing ghost symbols.

Lemma 4 and the commutation identities give an admissible weak-modal structure and full modal PA before any soundness theorem is invoked. Reflection is automatic, also at ghost heads.

Lemma 11 (truth of contextual collapse). Every admitted contextual-collapse generator is true in M before stage soundness is assumed.

Proof. At the changed atom, the two cut predicates agree by Lemma 7, and the argument is unchanged. Through a live unindexed head, source erasure makes the raw query identical. Through a ghost head, both tests are false. Through an indexed outer head, the smaller-context equivalence is itself a contextual-collapse generator for that holder below the outer label. It identifies the two raw proof tests there. Induction through the smaller context identifies their truth conjuncts. The whole support of that equivalence is below the outer label by global nesting. Boolean and quantified contexts follow by induction, at every assignment. $\square$

We next reconstruct a genuine original-language filtered structure from one main recorded argument $l\pi_H\phi$. This is the mechanism that turns external agents into parameters we can safely use in a lower genericity call.

Let D be a set of nonactive, nonexternal names retained in full. In the extension application it is $\{*\}$ together with a whole lower ideal. On original closed formulas define Q as follows:

1. Keep $Q_d = P_d$ in full for $d \in D$.
2. For active a , retain just its source argument shapes occurring in ϕ . On a sentence of such a shape, test its induced recorded lift in the cut below the assigned label; reject all other shapes.
3. For $e \in E$, retain P_e only on the argument shapes whose e -head is live in the main argument. Reject ghost-selected and other shapes.
4. Every other raw predicate is empty.

The source-shared label and mode tables make the second and third clauses unambiguous. The retained classes include *all* term variants and numerical instances of a shape. They do not keep merely finitely many accepted sentences. Put $N = F_Q$.

Lemma 12 (reconstruction). For the retained templates and all their term variants and assignments,

$$N \models \theta[s] \iff M \models l\pi_H\theta[s]. \quad (12)$$

Suppose in addition that the following three conditions hold:

- (S) Every used active cut is sound in M .
- (E) Every old P_e , $e \in E$, is fully sound in the original filtered structure F .
- (X) Under every retained live external argument, all active tests equal their full original source tests for every term variant and numerical assignment.

Then every changed predicate Q outside D is fully sound in N . The reconstructed family is admissible and recursively enumerable.

Proof. First prove (12) by formula induction, without assuming soundness. At an active atom the raw tests agree by the definition of Q . At a fully retained head they agree by erasure. At a live external head its old raw test has been retained. At a ghost-selected head both the original restricted test and the auxiliary test are empty. Other named heads are empty in both structures. Induction identifies the truth conjuncts, the Boolean connectives and all quantified instances. Keeping all variants is essential here.

For full soundness, an active acceptance is true in M by (S), and then true in N by agreement. An old external acceptance χ is true in F by (E). Heredity makes its recorded subtree Z -free. Condition (X) identifies all active raw tests in that subtree; all other occurring live tests are the same original tests. Formula induction therefore transfers F -truth of χ to M -truth of its lift, and (12) transfers it to N . All remaining predicates accept nothing. Thus no unrestricted old external predicate is left behind with unchecked assertions in N .

Finite shape recognition and recorded lifting are recursive. Cut consequence and the old predicates are recursively enumerable, so their restrictions are recursively enumerable and alphabetically invariant. When D is a downward ideal, enumerate its full predicates by positive finite path witnesses and dovetail their theorem enumerations. There is no need to decide the complement of D . The stipulated disjointness makes this enumeration compatible with the finitely many changed predicates. $\square$

The guard rule supplies (X). Inside a retained $K_e\chi$, an active $K_b^\nu\beta$ has a $G_{r(\epsilon\chi)}(\nu)$ guard, with coherent argument support below ν . Its argument rank is smaller than that guard rank. Lemma 9 therefore identifies its test with the original source test at this fixed label. The identification holds for every variant and assignment. Lemma 10 supplies the same identification for Z -free cofinal lifts.

Reducing an infinite external policy to a finite one

There is a useful preliminary restriction that does not involve labels. For a fixed theorem θ , keep the full raw predicates that an application requires to remain full, and restrict every outside predicate to its argument shapes occurring in θ . Hereditary closure is automatic when all subformula shapes are included. Formula induction gives agreement with the original structure on every retained argument variant, at all assignments. If the old outside predicates were fully sound, their restrictions remain fully sound. Only finitely many outside names can now be nonempty. This allows the genericity proofs to work with finite E while their conclusions apply to arbitrary admissible external families.

7 The common theory is generic over sound externals

The word generic here has a specific semantic meaning: the core remains sound when other agents are replaced by any admissible raw predicates already fully sound in the same filtered structure. It does not assert forcing genericity.

Lemma 13 (core genericity). Let F retain the full raw predicate $\text{Th}(U)$ at $*$. If every original external predicate is admissible and fully sound in F , then $F \models U$.

Proof. First assume only finitely many external predicates, with names in E , are nonempty. Form the core presentation with active set $\{*\}$. Its staged model uses the old external predicates on source erasures and indexed core cuts; ghost and other external predicates are empty.

Verify generated sentences in increasing order of whole support. Start with empty support. At a label ν , every smaller cut is sound because all its generated premises have already been verified. PA and Assigned Validity are true by model admissibility. Own validity holds because a valid argument of support below its head is an assigned-validity consequence in the corresponding cut. Own deduction uses closure of a single cut and the truth conjunct. Introspection uses Lemma 8 for raw membership at its later label, together with truth of its earlier atom. Reflection is built into the filter, and contextual collapse holds by Lemma 11. Markers are valid and belong to the indicated holder cuts.

Consider a main mechanicalness image. If its target is an indexed core head $K_*^\nu \chi$, the argument support is below ν . Soundness of that smaller cut makes the truth filter redundant in its column. The column is exactly its recursively enumerable cut proof column. This remains true for each assignment of the other parameters, so the existential index exists at every such assignment.

If the target is a live old $K_e \chi$, the guard rule includes this *outer* head. It supplies (X) throughout χ by Lemma 9. Heredity makes χ Z -free. Old soundness in F and equality of the tests on its subtree transfer every old accepted argument to a true staged argument. Equivalently, apply Lemma 12 to the whole recorded modal atom, retaining its outer live shape as well, with $D = \emptyset$. Its staged column is thus the recursively enumerable old predicate column. A ghost or empty external target has the empty column. This verifies all main mechanicalness images. No particular codes occur in the core presentation.

Finally, for an own prefix at ν , the body's whole support is below ν . Its truth has already been established, and it is a generated premise in that cut. The raw test and truth conjunct therefore both hold. Repeated prefixes are handled in the same support order. Padding makes every invoked guard point part of the support used in this induction. This proves all stage generators and hence their consequences sound.

For any U -theorem take a Z -free lift on C . It is a stage consequence by Lemma 10 and so true in the staged model. The same lemma identifies every indexed raw test with its full U test; unindexed tests agree by erasure. Formula induction returns the theorem's truth to F . This proves the finite-policy case without an assumption that U was already sound.

For an arbitrary external family, fix a U -theorem and apply the finite shape restriction above, retaining full $\text{Th}(U)$. Its outside predicates remain fully sound, and its truth agrees with that in F . The finite-policy case and transfer give the result. $\square$

8 Genericity for whole downward ideals

For a finite nonempty R^+ -antichain J , let

$$D_R(J) = \{i : i = a \text{ or } iR^+a \text{ for some } a \in J\}.$$

This ideal can be infinite. It is recursively enumerable by positive path witnesses. For $i \in \mathbb{N}$, define the metatheoretic rank

$$\rho(i) = \sup_{jRi} (\rho(j) + 1). \quad (13)$$

Well-foundedness defines ρ , and iR^+a implies $\rho(i) < \rho(a)$. No algorithm for these ranks will be used.

Lemma 14 (whole-ideal genericity). Let F retain the full raw predicates $\text{Th}(U)$ and $\text{Th}(T_i)$ for every $i \in D_R(J)$. At all other original names let the predicates be admissible and fully sound in F . Then F models U and every T_i for $i \in D_R(J)$.

Proof. Induct on $h(J) = \max_{a \in J} \rho(a)$, for all finite nonempty antichains. The induction assertion includes every member of the ideal, not only the frontier J . The finite-policy reduction permits us to assume that only finitely many outside predicates, with names in E , are nonempty. Retain the core and the *entire* ideal unchanged in that reduction.

Form the joint extension presentations with active set J . The staged model keeps the full core and full strictly lower original predicates, uses cut predicates at the labelled J -heads, and the old E -predicates on source erasure. Other outsiders and ghosts are empty. Verify all holder generators simultaneously in increasing whole support, first empty support and then successive labels. This is the inner induction. The structural verification is exactly the one in Lemma 13: it uses own cut closure, local prefixes, filtered reflection, modal PA and already established collapse truth. There are no mechanicalness main generators here. We must verify the particular codes.

Fix a main code image with source ϕ and owner w , where $w = *$ or wRa for its active holder a . Collect all original names in ϕ that lie strictly below J , and also w if it is original. Let A' be the R^+ -maximal members of this finite collection, and retain

$$D = \{*\} \cup D_R(A'), \quad D_R(\emptyset) = \emptyset.$$

Every member of A' is strictly below a member of J , so if $A' \neq \emptyset$,

$$\max_{b \in A'} \rho(b) < \max_{a \in J} \rho(a). \quad (14)$$

This is a finite semantic selection for the proof. It is not a selector in the source or stage enumeration. Every lower name occurring in ϕ is covered by the retained ideal. Lower predicates not mentioned there and outside D may be emptied for this local reconstruction.

Apply Lemma 12 to the recorded argument. Condition (S) holds because all active tests in this main generator use cuts below its current support maximum. Condition (E) is the old outside soundness hypothesis. Every retained live external argument is hereditary and Z -free; its guards supply (X) at fixed labels by Lemma 9. Consequently the reconstructed original structure N retains the full core and all the lower theories in D , and all predicates outside D are fully sound there. The source argument is true in N exactly when its recorded lift is true in the staged model.

If A' is nonempty, the outer induction hypothesis applied to A' makes U and the whole retained lower ideal sound in N . It includes the code owner w , even when w is not maximal in A' . If A' is empty, w must be the core, and Lemma 13 makes U sound in N because every original predicate there is a sound mask. Thus, for every n , an original w -proof of $\phi(\bar{n})$ implies truth of its recorded lift at n in the staged model. The code head's raw query is precisely this full original proof test, by source erasure. Equations (3) and (5) therefore verify both directions of the code: the truth filter

disappears on this argument, and the resulting test equals the displayed program's enumeration column.

The same reasoning covers empty-support codes. There are then no active cut or transport requirements; the lower ideal or core call verifies their source arguments. Padding is discharged by true markers. Own prefixes are true because their whole bodies precede their new labels and are already true and provable at the lower cuts. This completes the inner support induction. All active stage consequences and cut predicates are now sound.

It remains to recover the *whole* current ideal. A theorem of T_a , $a \in J$, has a Z -free cofinal lift provable in its stage by Lemma 10. Stage soundness makes it true. Cofinal recovery identifies its active tests with the full original tests; every unindexed query was already the original query on erasure. Formula induction returns its truth to F . Thus all frontier theories are sound in F .

Now take $i \in D_R(J)$ with $i \notin J$ and a T_i -theorem θ . Choose a coherent Z -free lift on C . Collect i and all original names in θ strictly below J , and take their finite maximal antichain A' . It satisfies (14). Reconstruct this lift retaining $\{*\} \cup D_R(A')$ in full. Stage soundness supplies (S), cofinal recovery supplies (X), and old outside soundness supplies (E). The masks outside the retained ideal are therefore fully sound. The lower induction hypothesis makes the unchanged full T_i -theorem true in N . Reconstruction and cofinal agreement transfer its truth to F . This argument works separately for every i and every theorem, even when the current ideal is infinite.

All original predicates, inside and outside the ideal, are now fully sound in F . Apply Lemma 13 at this point to obtain $F \models U$. This closes the outer induction and proves the finite-policy assertion. The preliminary finite shape restriction then gives the assertion for arbitrary outside families. $\square$

The proof order prevents two possible circles. Core genericity was proved independently of extension soundness. In an extension code verification, the outer call retains only a strictly lower original ideal, while the inner call uses smaller active cuts. Superior and incomparable side agents become fully sound restricted parameters; the proof never pretends that they are subordinates.

9 The actual family and its known schemas

Let F_{actual} be the filtered structure with full raw $\text{Th}(U)$ and every full raw $\text{Th}(T_i)$. At this point we have not assumed any of these predicates sound.

Lemma 15 (actual-family soundness). F_{actual} models U and every T_i . Its filters are redundant.

Proof. Fix a T_i -theorem θ . Collect i and all original modal names occurring in θ , and let J be the R^+ -maximal members of this finite set. It is a nonempty antichain and its ideal contains every collected name. Form F_J retaining the full core and full ideal predicates, with empty outsiders. Empty predicates are fully sound without a hypothesis about F_{actual} . Lemma 14 therefore gives $F_J \models \theta$.

Every raw query occurring in θ has been retained. Formula induction gives F_J/F_{actual} agreement on θ , so $F_{\text{actual}} \models \theta$. Arithmetic program numerals are unchanged, even if they encode formulas naming other agents. We have changed semantic predicates, not theories or enumerations. Since i, θ were arbitrary, all original theory predicates are fully sound. Lemma 13 now gives core soundness in F_{actual} . Full soundness removes every truth conjunct in (6). $\square$

Lemma 16 (effective core inheritance). For every original i and closed sentence σ , a U -proof of σ yields effectively a T_i -proof of σ and every finite own prefix of σ .

Proof. Put $\psi(x) = \sigma \wedge x = x$. A U -proof of σ gives one of $\psi(\bar{0})$. By (3), the concrete Σ_1 enumeration assertion $0 \in W_{d(\psi)}$ is true. Its finite computation trace supplies a PA proof. The particular core code in T_i then gives $K_*(\sigma \wedge 0 = 0)$; core reflection gives σ . Use Lemma 3 to obtain its own prefixes.

The numerical instantiation of the universal code requires care in the weak logic. Use the valid variable-instantiation implication from $\forall x A(x)$ to $A(y)$ with fresh y , then take its Assigned Validity instance at $y = 0$. This yields the required numerical instance without an unrestricted arbitrary-term substitution rule inside K_* . The finite trace is external evidence for this one computation, not an internal proof of the compiler's general correctness. $\square$

We can now verify the three assertions of Theorem 1. Full mutual truthfulness is a generator of every T_i , and its own prefix is also a theorem. Actual-family soundness makes that prefix true, giving the first schema. Modal PA and the structural generators and their prefixes similarly give self-doubting Epistemic Arithmetic.

For jRi , the particular sentence

$$\forall x (K_j \phi \leftrightarrow x \in W_{c(q,j,\phi)})$$

and its K_i -prefix are T_i -theorems. In the actual consequence model, assign $e = c(q, j, \phi)$ to obtain

$$\exists e K_i \forall x (K_j \phi \leftrightarrow x \in W_e).$$

This is numeral-assignment semantics, so no illegitimate modal term-generalization is used. Finally every $MC_j(\phi)$ is a closed generator of U . Lemma 16 gives it and its own K_i -prefix in every T_i , hence

$$K_i \exists e \forall x (K_j \phi \leftrightarrow x \in W_e).$$

The common core also contains parameter versions with universal closure outside the existential, and the same transfer applies to their closed sentences. All these formulas use the same fixed family.

Removing the temporary core

For each original i enumerate

$$T_i^{\text{orig}} = \{\sigma \in \mathcal{L} : T_i \vdash \sigma\}. \quad (15)$$

This is uniformly recursively enumerable, since original-language syntax is decidable. On original sentences, consequence from T_i^{orig} equals consequence from T_i . One implication composes the finite T_i -proofs of the premises. The other follows because a T_i -theorem in the original language is itself a premise in (15). Original-language validity is unchanged by the expansion: every original weak-modal structure can be extended by an arbitrary admissible interpretation for the extra operator.

The reduct of the sound actual structure therefore has exactly the knowledge predicates of T_i^{orig} . The required original EA, truthfulness, code and mechanicalness schemas survive. The code numerals still enumerate the right original argument columns, because those consequence columns have just been shown identical. The auxiliary ghosts and labels were never original axioms. Deleting K_* by this reduct leaves precisely $K_0, K_1, \dots$. This proves Theorem 1 once the ordinal input below is supplied. $\square$

10 Extracting the effective ordinal structure

The modal argument uses finite effective label procedures, not just the existence of large ordinals. This section supplies one recursive domain and total algorithms for both resemblance relations.

It also accounts for the partial conversions occurring inside the tracking algorithms. We use the relative term procedures and structural theorems as published mathematical inputs in Wilken [2]; their numbering below refers to arXiv version 5. They are not additional hypotheses on R or on the modal family.

10.1 Pure resemblance, reflection, and separated room

Wilken's pure structure is

$$(\text{Ord}; <, \leq_1, \leq_2), \quad \alpha \leq_i \beta \iff (\alpha; <, \leq_1, \leq_2) \prec_{\Sigma_i} (\beta; <, \leq_1, \leq_2),$$

with the relations defined simultaneously by induction on the upper endpoint. Thus $\leq_1$ already reflects in the language containing $\leq_2$. Ordinal addition and collapsing functions below are notation machinery; they are not added to the reflected relational language. We do not identify this pure structure with an arithmetic-expanded or modified finite-covering relation.

Definition 1.5 constructs the endpoints v_r . Its finite levels are covered by the relative systems T_{v_r} , starting with T_1 at level zero. Theorem 1.8 gives the actual cofinal chain $v_2 <_2 v_3 <_2 \dots$ below v_ω . The upper endpoint v_ω is excluded. In particular, v_1 has no strict $<_2$ successor and is not the initial chain point.

For $\alpha <_1 \beta$, encode the full finite diagram of a finite support in one existential formula, with parameters for the part already below α . Include order and both positive and negative resemblance facts, and the desired lower bound. Σ_1 -reflection supplies an exact copy below α fixing those parameters; the equality case is the identity. This is the finite reflection used in Lemma 7. Proposition 1.6 and the following interval property supply connectedness.

For the stronger room statement, suppose

$$\lambda = g_0 <_2 \dots <_2 g_k = \zeta, \quad k \geq 1,$$

and put $Y = \{g_0, \dots, g_{k-1}\}$. Lemma 1.7(2) reflects Y below λ , above any prescribed $u < \lambda$ and fixed finite $X < \lambda$, with every copied point $<_1$ -related to λ . Every original point of Y was $<_1$ -related to ζ , so the lemma applies to the whole copy, including its last point. Repeat with the next bound equal to the maximum of the preceding copy. This gives $X < Y_0 < \dots < Y_{m-1} < \lambda$ with whole-copy separation. If the fixed goal points are also $\leq_1 \lambda$, connectedness makes their union with all copies one coherent $\leq_1$ -chain. Initial subchains of each copy give every required G_r with $r \leq k-1$. These are exactly the room quantifiers in Lemma 9. Once the relation algorithms are available, a finite tuple search finds such copies; their existence proves search termination.

10.2 One public recursive domain

Set

$$B_0 = [0, v_1), \quad B_r = [v_r, v_{r+1}) \quad (r \geq 1).$$

A public code is a normalized finite T_1 term in B_0 , or a finite T_{v_r} term in B_r whose occurring parameters below v_r are fully written public trees from earlier blocks. The endpoint v_r has a finite tag and its native base term $\vartheta^{v_r}(0)$; when used in a later block it is an earlier parameter. This introduces no self-reference.

Definition 2.17, Theorem 2.18, and the paragraph after Corollary 2.19 give one uniform relative grammar and comparison procedure, with supplied parameter data. The largest written collapsing

index selects a sufficient finite fragment (Definition 2.21). Finite subterm sets and the comparison criterion of Lemma 2.26 determine the normal-form and countable-sort tests. There is no search through semantic hulls. The formal Ω_j symbols are finite grammar sorts, not input oracles for uncountable cardinals.

Calls to parameter data strictly decrease the finite block tag. Within a block, the relative procedures recurse on finite syntax. Carry additive and multiplicative normal forms, ordinal arithmetic, and principal-number classification with the earlier-block data; a bare recursive ordering of parameters would not suffice. Epsilon membership uses Lemma 2.29, fixed-point comparisons use Lemma 2.30, and logarithm and exponentiation use Lemmas 2.41 and 2.51. The epsilon endpoints are closed under the finite arithmetic operations used here. Results below a base become earlier parameter codes; other results have the normalized code in their containing block. An exact division in a guaranteed branch can alternatively be found by enumerating public codes and testing the witnessed multiplication equation. Existence at its caller makes that search terminate.

If hereditary spellings have duplicates, first allow recursive raw codes with recursive ordinal equality. Keep exactly those raw codes having no numerically smaller equal valid raw code. This is a finite recursive check. Each equality class has a least numerical code; converting a computed output checks only codes up to that output. Thus uniqueness need not be a separate assumption.

Coverage of each block follows from Theorem 2.18 and Definition 1.5. The same procedure works for every finite block tag, so this is one recursive presentation of $[0, v_\omega)$, not a nonuniform union of separately recursive orders. Arithmetic and comparison are constructed before tracking or resemblance is used. On this domain the limit-segmentation parameter in the source is zero, and no nonzero limit-index endpoint has to be coded.

10.3 Typed local bases and the conversion invariant

Tracking procedures also use temporary proper epsilon bases. Retain the public code of each ordinal while carrying, when needed, a native T_b representation with publicly coded parameters below b . A native term is not silently a fixed-endpoint public term. Every conversion has an explicitly checked source context and bound.

Let v be the public block base, and let a be a proper epsilon in that block. Use its relative successor bound a^+ from Convention 2.28. For countable inputs, the star-subterm estimate of Lemma 2.26 and Remark 2.27 makes

$$\xi < a^+ \implies \text{the public/native conversions between the } v \text{ and } a \text{ contexts are defined on } \xi. \quad (16)$$

These are the partial translations on page 17. Temporary high-sort expressions instead carry the actual star-subterm restriction in Definition 2.45. The routine $\iota_{v,a}$ acts on admitted parts of the collapsing argument whose star bounds are below a .

The important outputs obey

$$\lambda_a < a^+, \quad \mu_a < a^+, \quad x \leq \mu_a \implies \rho_x^a \leq \lambda_a < a^+. \quad (17)$$

The references are Lemmas 2.48, 3.7 and 3.14(b). Thus these native results return to their public context. For consecutive proper bases of a μ -covered reference sequence, $a_{i+1} \leq \mu_{a_i} < a_i^+$. Lemmas 2.48(d), 3.8, 3.26 and the nested localization bounds in Lemma 2.33 identify the suppressed relative contexts on their overlaps.

To resolve a parameter p with $a_i \leq p < a_{i+1}$, use

$$p < a_{i+1} \leq \mu_{a_i} < a_i^+$$

to convert its existing public payload into T_{a_i} . The case $p = a_i$ is the base term. Parameters below a_i move to an earlier reference position. After endpoint positions are reached, calls decrease the public finite block tag. This is an effective decreasing resolution order, and produces the actual native resolved syntax of Definitions 3.39–3.41. Its natural-valued length is computable; it is not the length of some different public normalization.

Endpoint bases have explicit clauses in Definition 3.15: their λ/μ output is the next finite endpoint. Interior inputs already have the public native block representation; equality with the next endpoint is dispatched before a proper-base conversion. Zero has its own $\kappa, \text{dp}, \text{tc}$ cases. No caller asks for λ_0 or μ_0 .

10.4 Tracking sequences and finite evaluation

Here is the procedure order and the conversion information needed by its callers. The underlying finite operations are Wilken’s published relative procedures; the preceding invariant puts them on the one public datatype.

Localization and fine localization process finite principal-subterm sets (Definitions 2.32, 2.34, 2.35 and 2.38). Fixed-point tests use Lemma 2.30 rather than semantic supremum searches. The terminating bar operations are justified in Lemmas 2.35–2.40. The χ and ζ data are finite normal-form operations (Definitions 3.3 and 2.42); λ and μ use the translations with the bounds in (17). A ρ caller has $x \leq \mu_b$, so its output has the same justified return conversion.

Tracking-sequence assignment ts uses finite localization recursion for multiplicative principal inputs (Definition 3.19, Lemma 3.21), and finite multiplicative decomposition for other principal inputs (Definition 3.20). A temporary call $\text{ts}^{a_i}(\beta)$ has the source bound $\beta \leq a_n$; nested localization puts it below a_i^+ and permits its conversion. Endpoint extension is the finite prefix procedure of Definition 3.24. The covering procedures of Definitions 3.25–3.28 select finite localization subsequences.

Two evaluation details matter for effectiveness. The bound $\hat{\gamma}$ in Definition 3.30 is introduced as a semantic minimum, but is not computed by a least-ordinal search. At an actual h -call in o , the input has $\beta \leq \gamma$ or $\beta \leq \mu_\gamma$. Since $\hat{\gamma}$ exceeds both, the advertised h -domain follows, and the computable conversion bound is $\beta < \gamma^+$. The evaluator can branch on the ordinary comparisons with γ and μ_γ .

Also, the overlined operation in Definition 3.33 truncates factors strictly below β from the multiplicative normal form of μ_δ . Its output is at most μ_δ . It is not the ordinary product $\mu_\delta \cdot \beta$. The actual truncation preserves the covered next entry and the decreasing native-height bound of Lemma 3.7. Thus sk and h are finite typed operations. Evaluation o then uses the finite sequence-length descent of Definition 3.35 and Theorem 3.36, after the endpoint fixed-point cases. Final sums and products are formed on public codes. A large evaluated product is not passed through a local partial translation merely because its factors had native types. Lemma 3.49 and Corollary 3.50 give finite containing blocks for these evaluations.

10.5 Scheduling the component procedures without a cycle

Definitions 3.37, 3.43 and 3.44 appear mutually dependent if read as a single informal program. They have the following finite construction order:

1. Define principal κ and principal ν at indices greater than 1 by the already constructed evaluator o . For nonempty contexts set $\nu_0 = o(\vec{\alpha})$ and $\nu_1 = o(\vec{\alpha}) \cdot 2$ directly. At a proper last base b , the respective native domains are bounded by λ_b and μ_b , so (17) licenses conversion.
2. Define general κ and dp by Definition 3.43, with principal ν already available.
3. Define nonprincipal ν by Definition 3.44, using the already total κ and dp procedures.

In the epsilon clause for dp, the call is $\nu_{\mu_b}^{\vec{\alpha} \frown b}$. The index μ_b is additive principal, being the stated ω -power or a finite endpoint, and is greater than 1. Thus this is a call to the *principal* evaluator from step 1, not to the later general ν recurrence. It satisfies the last-index restriction in Definition 3.18. No recursive cycle remains in this scheduling.

The measure for step 2 is lexicographic:

$$(\text{native resolved natural term length, reference-list length}). \quad (18)$$

Additive clauses use proper nonzero components. The non-epsilon dp clause uses the shorter resolved logarithm or divided-logarithm arguments of Lemma 3.42(2). The epsilon clause appends b and calls κ/dp at λ_b ; Lemma 3.42 supplies the strict native resolved-length decrease after that append. The same-value clause removes a last context base above the input: its fully resolved payload is unchanged and the reference-list length decreases. Zero, one, base and endpoint cases terminate directly.

The appended list must itself be covered. Its caller has $b \leq \lambda_a$ for the preceding base a , and Lemma 3.14(c) gives $b \leq \mu_a$. If $\mu_a < a$, then $\lambda_a < a^2$ and there is no new proper epsilon greater than a in that interval; otherwise the greatest epsilon below λ_a equals that below μ_a . Endpoint cases have the explicit next-endpoint bound. This verifies the reference-sequence hypothesis of the native length estimate.

Step 3 recurses on a successor predecessor or proper additive components. Its additional κ/dp calls at ρ -indices invoke procedures already made total; they do not require a claim that every ρ term is shorter than the original ν input. Lemma 3.14(b) supplies their local λ -domain. Equations (16)–(17) and the parameter-resolution procedure provide the native representations at each recursive call and public codes on return.

10.6 The relative-tracking caller bound

Definition 3.85 constructs $\text{ts}[\vec{\alpha}](z)$ by finding the largest position k with $o(\vec{\alpha} \upharpoonright k) \leq z$ in a finite list. The $k = 0$ case uses the public endpoint segment. For $k > 0$ it forms the public arithmetic value

$$z' = a_k \cdot ((1/o(\vec{\alpha} \upharpoonright k)) \cdot z). \quad (19)$$

The parenthesized expression denotes the source's exact ordinal quotient, not an ordinary real reciprocal. Its existence is Lemma 3.53. Before conversion to the native T_{a_k} context we need the actual bound $z' < a_k^+$; $z < v_\omega$ alone would not give it.

If k is not the last position, maximality gives $z < o(\vec{\alpha} \upharpoonright (k+1))$, and Lemma 3.49 bounds the latter by $o(\vec{\alpha} \upharpoonright k) \cdot \hat{a}_k$. At the last position the declared domain gives $z \leq \kappa_{\lambda_{a_k}}^{\vec{\alpha}} + \text{dp}^{\vec{\alpha}}(\lambda_{a_k})$; Corollary 3.50 bounds that expression strictly below $o(\vec{\alpha}) \cdot \hat{a}_k$. In both branches,

$$z < o(\vec{\alpha} \upharpoonright k) \cdot \hat{a}_k.$$

The exact quotient in (19) is therefore below $\hat{a}_k$. This bound is multiplicative principal and strictly greater than a_k . Closure under the product of two smaller factors gives

$$z' < \hat{a}_k \leq a_k^+. \quad (20)$$

The last comparison is stated following Definition 3.30. No computation of the semantic $\hat{a}_k$ is needed: it is used to prove a bound on the public arithmetic output. Equation (20) licenses its actual partial conversion. Lemmas 3.52–3.53 place the output tracking sequence in the required relative initial segment. If a_k is an endpoint, the native public endpoint block is used directly.

10.7 Tracking chains and total relation decisions

Index-chain and reference data are finite scans of publicly coded lists (Definitions 3.54–3.62 and 3.74). Reference pointers go to earlier index pairs, and Lemmas 3.63 and 3.75 give the component domains. Maximal extension uses the typed μ, λ , logarithm and ρ procedures (Definitions 3.64 and 3.71). Lemma 3.67 proves termination after contracting adjacent μ/λ pairs; its endpoint clause stops rather than iterating the next-endpoint operation indefinitely. Recognition of tracking chains uses shorter proper initial chains and finite extension lists (Definitions 3.68–3.69). Chain evaluation is a finite fold of the already total component procedures (Definition 3.76).

The tracking-chain assignment tc of Definition 3.87 uses ts for principal inputs and finite additive decomposition otherwise. Every relative tracking preparation has the bound just proved. Lemma 3.89 verifies both the chain condition and evaluation back to the input. The greatest branching index of Definition 4.5 recurses on proper prefixes. Consequently the reach calculation lh in Corollary 4.6 is effective on the one public domain. Its result is a public finite term or the separate sentinel ∞ , never an uncoded ordinal passed to an arithmetic procedure.

The closed-interval property now gives the total decision

$$\alpha \leq_1 \beta \iff \alpha \leq \beta \text{ and } (lh(\alpha) = \infty \text{ or } \beta \leq lh(\alpha)). \quad (21)$$

Endpoint equality is recursive on public codes; $lh(0) = 0$ and $lh(v_r) = \infty$ for $r \geq 1$ are explicit cases.

Theorem 4.2(b) computes the greatest strict $<_2$ predecessor p of a nonminimal β , apart from its stated nonzero limit-index endpoint exception. Such endpoints do not occur below v_ω . Minimality is a finite tracking test. If $\alpha <_2 \beta$, greatestness gives $\alpha \leq p$ in the ordinary order. Since $\alpha, p \leq_1 \beta$, connectedness gives $\alpha \leq_1 p \leq_1 \beta$. The pure respect property

$$u \leq_1 v \leq_1 w, \quad u \leq_2 w \implies u \leq_2 v$$

therefore gives $\alpha \leq_2 p$. Transitivity supplies the converse, so

$$\alpha <_2 \beta \iff \alpha \leq_2 p. \quad (22)$$

Test equality first, then ordinary order and minimality, and otherwise replace β by its computed greatest predecessor. Each iteration is a total finite procedure and strictly decreases the represented ordinal. Well-foundedness proves termination; a recursive running-time bound is unnecessary. Zero cannot conceal a real predecessor, because it has no strict $\leq_2$ successor. As endpoint checks, v_1 is minimal without strict successors, v_2 is minimal with strict successors, and v_n has predecessor v_{n-1} for $n \geq 3$.

Lemma 17 (effective label package). There is one recursive presentation of $[0, v_\omega)$ with recursive $<, \leq_1, \leq_2$ and all four properties in Section 3.

Proof. The public grammar, normalization and arithmetic precede tracking. Every temporary conversion has its public payload and the caller bounds above. Finite parameter resolution, the evaluation descent, the scheduled component recursion and finite chain operations return public codes and terminate. Equations (21) and (22) give total relation algorithms. Abstract reflection, separated room and the cofinal chain were established in the first subsection. Finite searches for guard copies use these relation algorithms, with termination supplied by room. No operation returns v_ω as a member code; the only unbounded reach is the explicitly separate ∞ sentinel. This supplies exactly the package used in the modal proof. $\square$

Scope of the argument

The proof above covers the complete 3-1-3 statement: arbitrary recursively enumerable well-founded R , unrestricted modal formulas, one uniform family, all three known schemas, and the original-language reduct. Its nontrivial published dependencies are Alexander’s weak modal completeness and arithmetic semantics, the ordinary recursion and parameter theorems, and Wilken’s relative ordinal procedures and pure resemblance theorems. The uniform extraction of those procedures is included rather than left as an extra effective-ordinal assumption.

This manuscript expands a complete proposed hand proof whose recorded modal and ordinal checks have been completed. Those checks do not constitute a proof-assistant certificate or external peer review. The separate Lean development still has unfinished dependencies and proofs. In particular, no unconditional Lean theorem for full 3-1-3 is asserted here. The source correspondence and exact document-build checks are recorded alongside this paper.

References

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